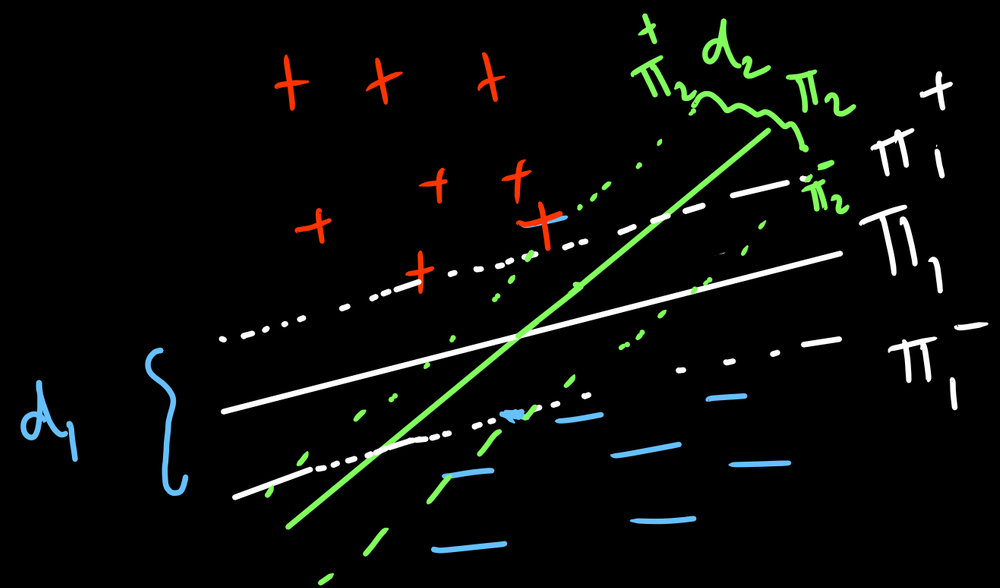
→ **SVM** (Support Vector Machine)

↳ Generally used for classification

↳ Popular in 2000's



$d_1 > d_2$

$\downarrow \qquad \downarrow$

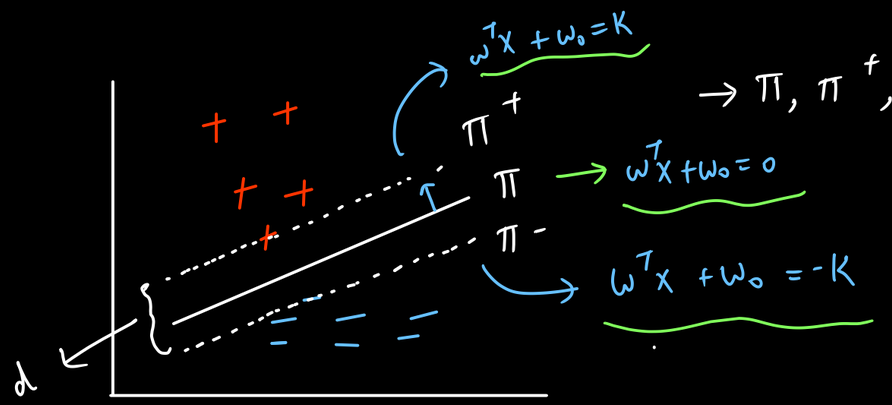
$1 \qquad 2$

π_1 is the best hyperplane

$\because d_1 > d_2$

→ Maximize the margins

SVM (Maximum margin Classifier)



$\rightarrow \pi, \pi^+, \pi^-$ are parallel

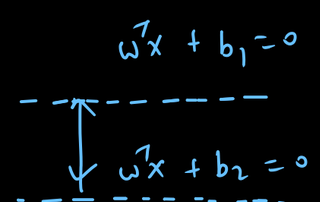
$$\text{Margin} = \text{Dist}(\pi^+, \pi^-)$$

$$\pi^+ \quad w^T x + w_0 = K \longrightarrow w^T x + \overbrace{w_0 - K}^{b_1} = 0$$

$$\pi^- \quad w^T x + w_0 = -K \longrightarrow w^T x + \overbrace{w_0 + K}^{b_2} = 0$$

$$= \frac{b_2 - b_1}{\|w\|}$$

$$\text{Margin} = \text{Dist}(\pi^+, \pi^-) = \frac{w_0 + K - (w_0 - K)}{\|w\|}$$



$$d = \frac{2K}{\|w\|}$$

$$\left\{ \frac{b_2 - b_1}{\|w\|} = \right.$$

$$\max(d) = \frac{2K}{\|w\|} = \frac{2}{\|w\|}$$

$$\boxed{\begin{aligned} &\max\left(\frac{2}{\|w\|}\right) = \text{Max. margin} \\ &\hookrightarrow \min \|w\|^2 = \text{Min Loss} \end{aligned}}$$

$\rightarrow$ Maximize margin = Min. Error

$w^T x + b = 0$

(1) $3x + 4y + 2 = 0 \longrightarrow d = \dfrac{2}{\|w\|}$

$= \dfrac{2}{\sqrt{3^2+4^2}} = \dfrac{2}{5} = 0.4$

$\rightarrow$ Error $= \|w\|^2 = 25$

(2) $6x + 8y + 2 = 0$

$\rightarrow d = \dfrac{2}{\|w\|} = \dfrac{2}{\sqrt{6^2+8^2}} = 0.2$

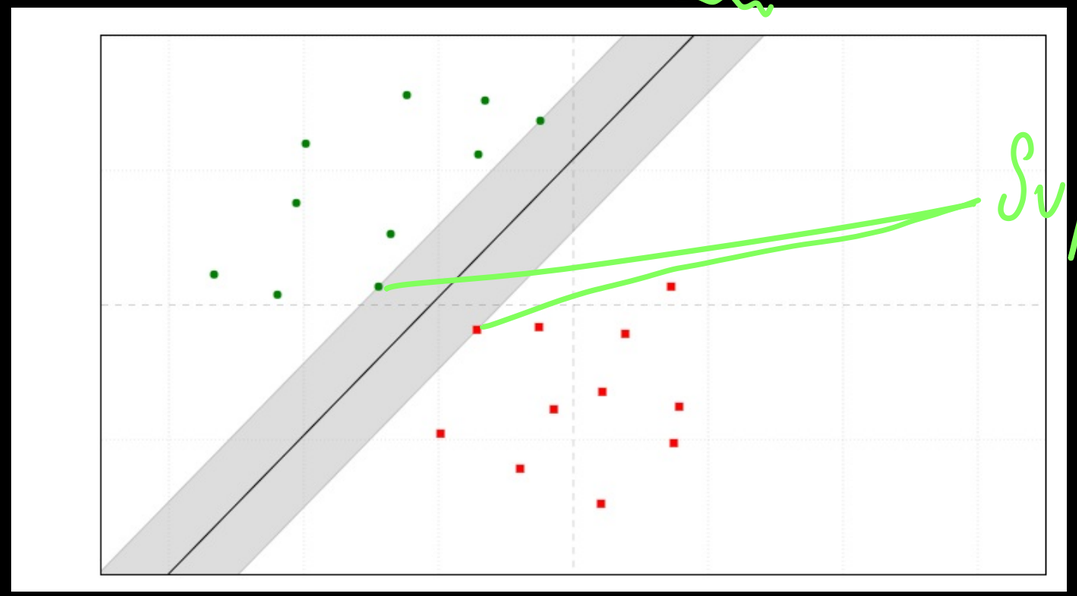
$\rightarrow$ Error $= \|w\|^2 = 6^2+8^2 = 100$

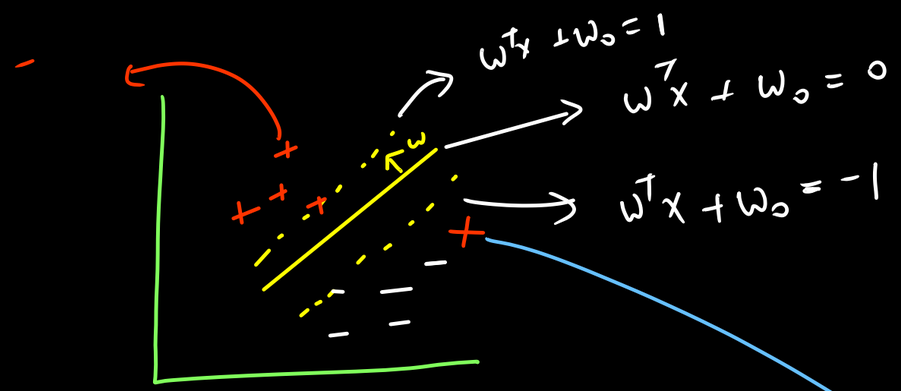
logistic

$y_i \rightarrow \begin{cases} 0 \\ 1 \end{cases}$

SVM

$y_i \rightarrow \begin{cases} -1 \\ +1 \end{cases}$





$\text{sign}(x) = 1 \quad x \geq 0$
$\quad\quad = -1 \quad x < 0$

$\text{sign}(w^T x + w_0)$

Red $\rightarrow$ +ve $\quad w^T x + w_0 > 0$

White $\rightarrow$ −ve $\quad w^T x + w_0 < 0$

$y_i = 1$	$y_i = -1$	$y_i = +1$
$w^T x_i + w_0 > 0$	$w^T x_i + w_0 < 0$	$w^T + w_0 < 0$
$\boxed{y_i (w^T x_i + w_0) > 0}$	$y_i (w^T x_i + w_0) > 0$	$y_i (w^T x_i + w_0) < 0$

$y_i (w^T x_i + w_0) \geq 0 \longrightarrow$ Correctly classified

$y_i (w^T x_i + w_0) < 0 \longrightarrow$ Incorrect

SVM (Hard margin)

$$\left\{\begin{array}{l} \max\left(\dfrac{2}{\|w\|}\right) \quad \text{s.t} \quad y_i\left(w^T x_i + b\right) \geq 1 \\ \updownarrow \\ \equiv \min \|w\|^2 \quad \text{s.t} \quad y_i\left(w^T x_i + b\right) \geq 1 \end{array}\right.$$

Hard Margin

$$L = \min_{w} \|w\|^2 \quad \text{s.t} \quad y_i\,(w^T x_i + b) \geq 1$$

Soft margin

$$L = \min_{w,\, \xi} \left(\|w\|^2 + C \sum \xi_i \right)$$

ξ_i → Error (Slack)

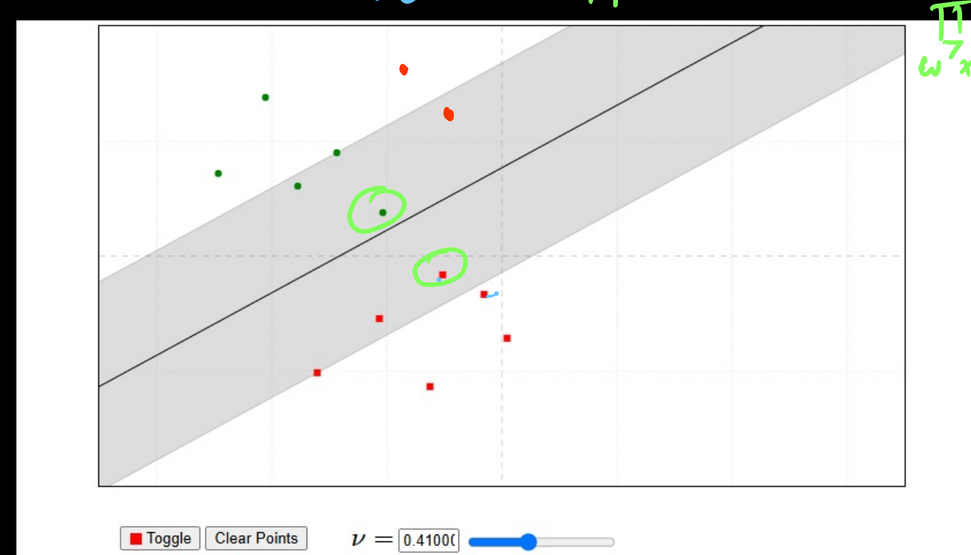
$\xi_i > 0$

$$\text{s.t} \quad y_i\,(w^T x_i + b) \geq 1 - \xi_i$$

$\pi^+ \rightarrow w^T x + w_0 = 1$

$\pi^- \quad w^T x + b = -1$

$\boxed{\text{Langen Margin}}$



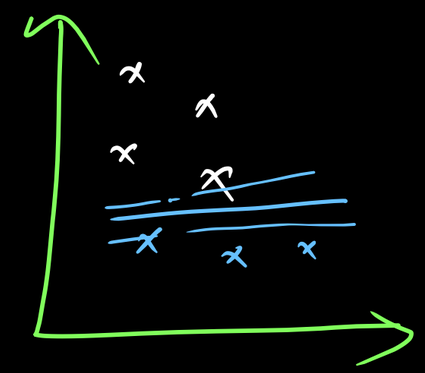
→ Break until 22:41

→ Hyperparameter (C)

$$L = \min_{w, \xi_i, b} \left(\|w\|^2 + C \sum \xi_i \right)$$

$\xi_i \geq 0$

s.t. $y_i (w^T x_i + b) \geq 1 - \xi_i$



1) $C \uparrow\uparrow$ $\xi_i \downarrow$ $\xi_i = 0$

$C = 10^4$ $\xi_i \approx 0$ → Hard margin → Overfit

↳ 10^{-8}

↳ Sensitive to training data

$C = 1 \downarrow$ $\xi_i \uparrow$ → High bias

↳ Underfitting

$$L = \min_{w,b,\xi} \left(\|w\|^2 + C \sum \xi_i \right)$$

$\xi_i > 0$

s.t. $y_i (w^T x_i + b) \geq 1 - \xi_i$

$\xi_i > 1 - y_i (w^T x_i + b)$

$\xi_i > 2$

$\xi_i > 0$

$\xi_i > 2$

$$= \boxed{\min_{w,b} \left(\|w\|^2 + C \sum \max(0, 1 - z_i) \right)}$$

$z_i = y_i (w^T x_i + b)$

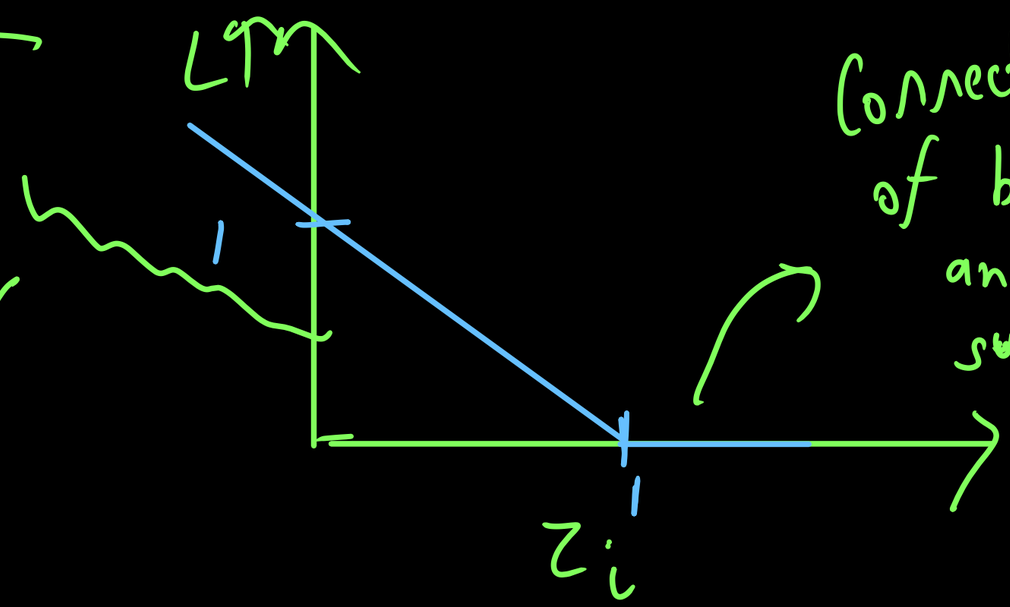
Hinge Loss

Hinge loss

$$= \sum_{i=1}^{N} \max(0, 1 - z_i)$$

$$z_i = y_i \cdot (w^T x_i + b)$$

$\max(0, 1 - z_i)$



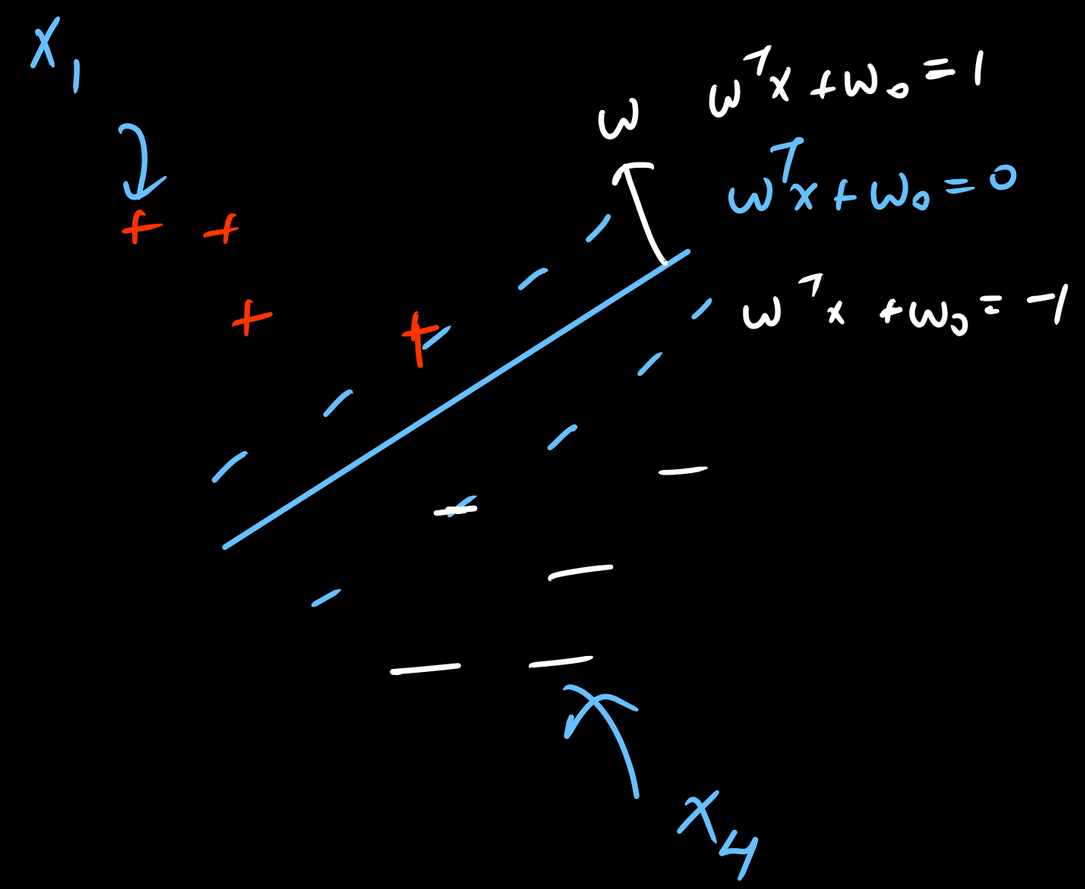
Incorrect side

Correct side of boundary and beyond support vector

$+ \rightarrow 1$

$- \rightarrow -1$

$$\text{sign}\left(w^T x + b\right)$$



$x_1 \rightarrow \quad w^T x_1 + b > 1$

$y_i \rightarrow 1$

$$\underbrace{y_1 (w^T x_1 + b)}_{z_i} > 1$$

3

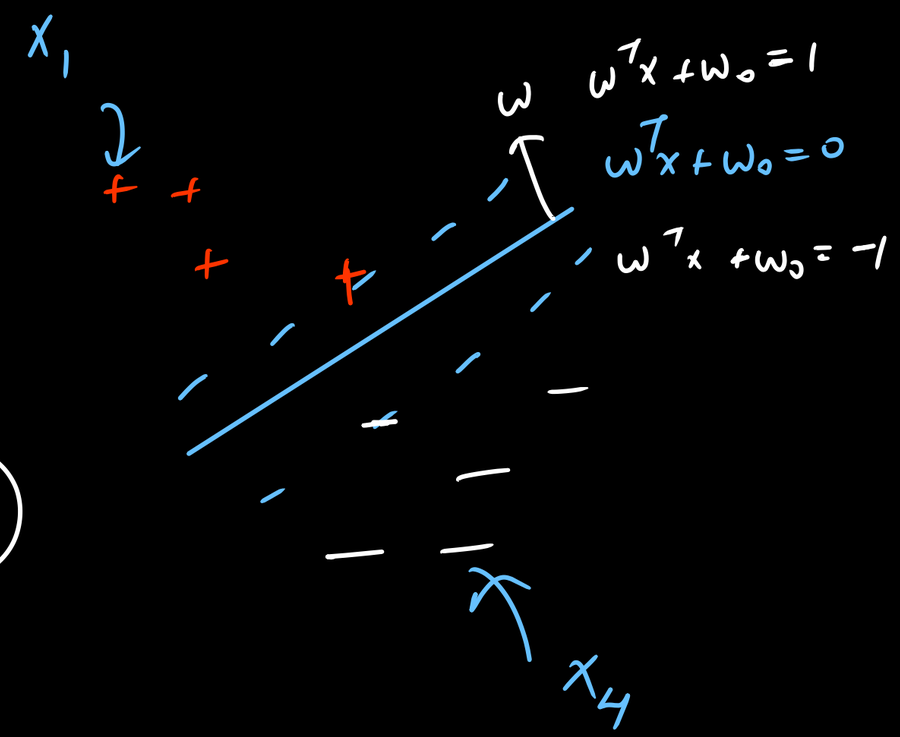
✓ Error $= 0$

✓ Error $= \max(0, 1-3) = 0$

$+ \rightarrow 1$

$- \rightarrow -1$

$\text{sign}(w^T x + b)$



$x_1 \rightarrow \quad w^T x_4 + b < -1 \quad = -3$

$y_4 \rightarrow -1$

$\underbrace{y_1 (w^T x_1 + b)}_{z_i} > 1$

$z_i = -1 \cdot (-3) = 3$

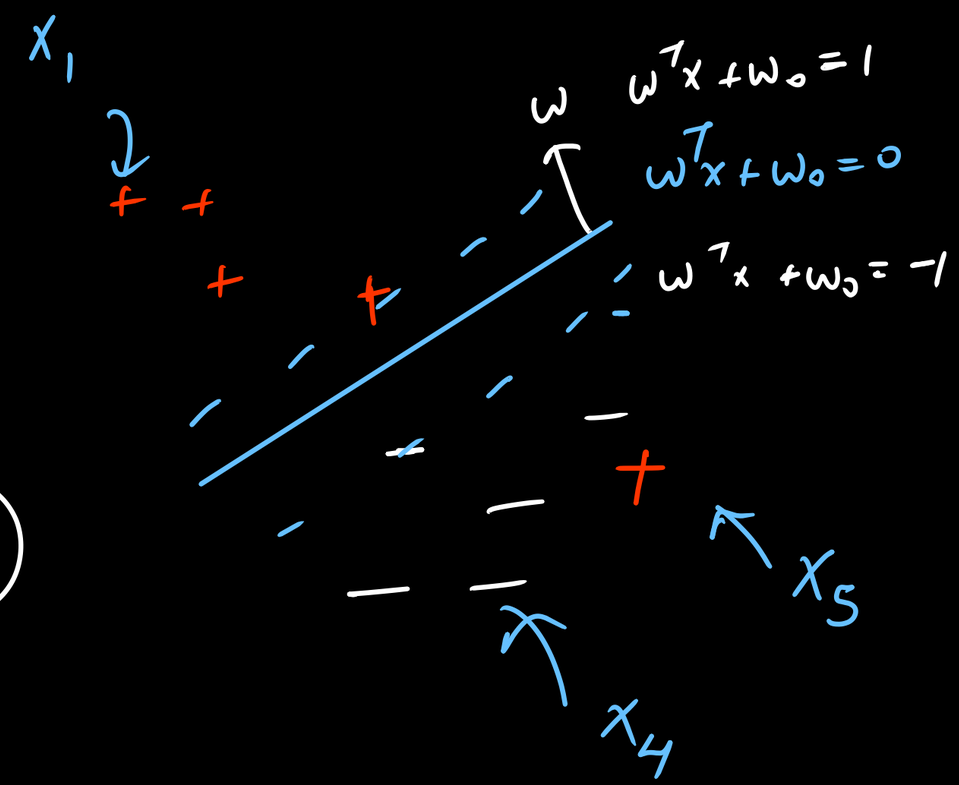
$L = \max(0, 1 - z_i)$

$= \max(0, 1 - 3) = 0$

$+ \rightarrow 1$

$- \rightarrow -1$

$\text{sign}(w^T x + b)$



$x_1 \rightarrow$ $w^T x_5 + b < -1$ $= -3$

$y_5 \rightarrow +1$

$z_i = \underbrace{y_1 (w^T x_1 + b)}$ < -1

$= -3 \times 1 = -3$

$L = \max(0, 1 - z_i) = \max(0, 1+3) = 4$

Logistic Regression

$$-\sum y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)$$

$$+ \lambda \|w\|^2$$

$y_i \to 0$

$y_i \to +1$

$\lambda = 0 \to$ Overfit

$\lambda \uparrow \to$ Underfit

SVM loss

$$\sum \max(0, 1 - z_i)$$

$$z_i = y_i(w^T x_i + b)$$

$$+ \frac{1}{C}\|w\|^2$$

$y_i \to +1$

$y_i \to -1$

$C \simeq 0 \to$ Underfit

$C \uparrow \to$ Overfit

$$\boxed{C = \frac{1}{\lambda}}$$